Phan Trong Dat

COMPUTER SCIENCE & ARTIFICIAL INTELLIGENCE

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Summary

Computer Science student with hands-on experience in **NLP, topic modeling, and computer vision**. Experienced in building practical AI systems, from research-oriented experimentation (published at EMNLP 2026) to production-grade data pipelines and IoT applications.

Education

Hanoi University of Science and Technology
Bachelor of Science in Computer Science *2023 – Present*

- Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Science, Algorithms & Data Structures, Artificial Intelligence, Database Systems

Vinh Phuc High School for the Gifted
Information Technology Specialist *Aug. 2020 – Jul. 2023*

Publications

LLM Feedback for Topic Model with Preference Optimization and Contrastive Learning 2026
EMNLP 2026 Main Conference – Co-author (Author #2) *[Paper]*

Experience

Software Engineer
GeoIT VNPT *Mar. 2026 – Jul. 2026*

- Develop AI-integrated geospatial systems, backend services, and data processing workflows for location-based applications.
- Design and implement scalable data pipelines for processing and analyzing geospatial data streams.

AI Intern
BA GPS *Jun. 2024 – Dec. 2024*

- Built computer vision modules for real-time object, face, and hand detection using Python, OpenCV, and YOLO-based pipelines.
- Optimized inference performance for deployment on embedded hardware with limited computational resources.

Robotics Developer & Mentor
Vietnam Robotics Challenge *2021 – 2022*

- Designed and programmed competition robots; mentored a student team that achieved runner-up performance nationally.
- Developed motion control logic and sensor integration for reliable autonomous robot operation under competition conditions.

Projects

AdverTest – Adversarial Robustness Testing for Perception Models
2026 – Present *(In Progress)*

- Developing a modular adversarial robustness testing platform for computer-vision perception models with plugin-based attacks, model/dataset adapters, and reproducible evaluation workflows.
- Built attack-dataset generation workflows covering gradient- and patch-based methods including FGSM, PGD, MI-FGSM, C&W, TOG, DAG, SAM2-PGD, DPatch, and Thys patch.

- Integrated anonymized dataset processing, AP/degradation evaluation, FastAPI/CLI execution, and KITTI/YOLO-oriented benchmark workflows; extending the system toward defense and retraining evaluation.
- Stack: Python, FastAPI, NumPy, PyTorch, Ultralytics/YOLO, pytest, Docker.

BrainFlow – Multi-Agent B2B Marketing Strategy Engine
2026 – Present *(In Progress)*

- Developing a deterministic multi-agent AI pipeline for B2B marketing strategy, spanning goal setting, situation audit, strategy formulation, budgeting, and cross-functional review.
- Combined LLM agents with Python-based math and budget interceptors for weighted SWOT, gap analysis, and budget enforcement to reduce numerical hallucination.
- Stack: Python, FastAPI, LangChain, Groq LLM, ChromaDB, Next.js, Tailwind CSS.

MedTutor – Multimodal RAG Copilot for Medical Education
2026 – Present *(In Progress)*

- Developing a multimodal Vietnamese medical-education RAG assistant with citation-grounded answers, safety guardrails, and ingestion for text, documents, images, audio, and video.
- Implemented seven retrieval strategies and an evaluation harness for hit@k, MRR, citation rate, and safety accuracy using BGE-M3, Qdrant, and Qwen3/Ollama.
- Stack: Python, FastAPI, Next.js, Qdrant, BM25, BGE-M3, Ollama, Docker.

PRISM – Mixed-Precision Quantization Pipeline for LLMs
2025 – Present *(In Progress)*

- Designed and implemented a low-cost, deployment-time calibration-free mixed-precision quantization pipeline targeting large language models (e.g., LLaMA-2-7B).
- Developed a meta-learner (Stage 0) trained on layer profiling features (shape, outlier statistics, RTN error) to predict optimal per-layer bit assignments.
- Implemented a data-free profiling stage with budgeted discrete one-hot bit assignment under realistic hardware memory constraints.
- Applied QUIC output-space correction using synthetic-only data, and integrated runtime assembly with Marlin-preference and PyTorch GEMM fallback.
- Stack: Python, PyTorch, CUDA.

LLMFeed – Neural Topic Modeling (EMNLP 2026) [Paper]
2025 – 2026

- Contributed to the implementation and experimental evaluation of LLMFeed, a neural topic model that maximizes mutual information and applies group-topic regularization.
- Ran and benchmarked competing approaches including LISA and TopicGPT, systematically comparing outputs across multiple coherence and diversity metrics.
- Extended codebase with additional dataset handlers, evaluation scripts, and training configurations to support ablation studies.
- Stack: Python, PyTorch, Sentence-Transformers, BERTopic.

AI Parking
2026

- Designed a production-like edge-first parking system with an async AI pipeline integrating YOLO (detection), OCR (license plate recognition),
- Architected a decoupled backend: the FastAPI service owns API, database (MySQL/SQLite via SQLAlchemy async), and WebSocket concerns, while an AI runtime handles frame routing and event production via bounded `asyncio.Queue` instances.
- Integrated FAISS vector search for Re-ID and built a real-time WebSocket dashboard for live event monitoring.
- Stack: Python, FastAPI, YOLO, FAISS, SQLAlchemy, asyncio, HTML.

Robotic Arm for People with Disabilities
Mar. 2022 – Jan. 2023

- Built a voice- and infrared-controlled robotic arm to assist people with physical disabilities in performing everyday tasks.
- Programmed motion control logic on Arduino and integrated hardware–software components for stable prototype operation.
- Stack: Arduino, C++, Infrared sensors, Voice recognition module.

Skills

Programming Languages: Python, JavaScript, C/C++, HTML, CSS
AI / ML Frameworks: PyTorch, NumPy, Pandas, Scikit-learn, Hugging Face Transformers, OpenCV, YOLO, BERTopic
Backend & Tools: FastAPI, Flask, Streamlit, SQLAlchemy, FAISS, MySQL, SQLite, Arduino
Developer Tools: Git/GitHub, VS Code, Jupyter Notebook, WandB, Linux
Research Areas: NLP, Topic Modeling, Computer Vision, Time-Series Forecasting, LLM Quantization, Edge AI
Foreign Languages: Vietnamese (Native), English (technical reading and writing)

Honors, Awards & Leadership

- **2nd Prize** – National Science & Technology Competition for Students (2023)
- **Top 10 & People's Choice Award** – Venture X 2026, Entrepreneurship Lab (E-Lab), VinUniversity
- **2nd Prize** – Provincial Youth Creativity Contest (2022)
- **Runner-up Team Mentor** – Vietnam Robotics Challenge (2022)
- **President** – Badminton Club (2022)